Power and resource tradeoffs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Distinguish supplied alternative-trial frequencies, per-study cost and unmeasured null error.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Pilot experiments

My remembered idea, example and correction

Privacy and data governance

My remembered idea, example and correction

Sensitivity analysis

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Power concerns rejection probability under a specified alternative model, test and design. Supplied main simulation counts are sixty-two of one hundred at n = 20 and eighty-three of one hundred at n = 40: 62% and 83% estimated detection frequencies. At three credits per independent unit, per-study costs are sixty and one hundred twenty credits.

  The descriptive difference is twenty-one percentage points for sixty extra credits per study. Separate picture has sixty and eighty detections out of one hundred at the same sample sizes, with unit cost two: 60% and 80%, costs forty and eighty resource units. Keep these scenarios separate. Finite simulation estimates are not exact known power, and no simulations are executed by the arithmetic code.

  The null false-positive rate is unmeasured. Effect size, variability, unit dependence and test assumptions matter; changing them needs new model-specific planning. Per-study cost is different from total hypothetical work over all simulation trials. No actual billing, pupil study or guaranteed performance is supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- For a fixed alternative 100 simulations detect 62 times at n 20 and83 at n 40. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Main quantity	n = 20	n = 40
Supplied detections / alternative trials	62/100	83/100
Estimated detection frequency	62%	83%
Independent units per study	20	40
Per-unit cost	3 credits	3 credits
Per-study cost	20 × 3 = 60 credits	40 × 3 = 120 credits

Main quantity	n = 20	n = 40
Large minus small	21 percentage points more observed detection	60 extra credits per study
Null error and actual-study guarantee	Not supplied	Not supplied

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Power concerns rejection probability under a specified alternative model, test and design. Supplied main simulation counts are sixty-two of one hundred at n = 20 and eighty-three of one hundred at n = 40: 62% and 83% estimated detection frequencies. At three credits per independent unit, per-study costs are sixty and one hundred twenty credits.

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  The null false-positive rate is unmeasured. Effect size, variability, unit dependence and test assumptions matter; changing them needs new model-specific planning. Per-study cost is different from total hypothetical work over all simulation trials. No actual billing, pupil study or guaranteed performance is supplied.

- A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Power concerns rejection probability under a specified alternative model, test and design. Supplied main simulation counts are sixty-two of one hundred at n = 20 and eighty-three of one hundred at n = 40: 62% and 83% estimated detection frequencies. At three credits per independent unit, per-study costs are sixty and one hundred twenty credits.

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  The null false-positive rate is unmeasured. Effect size, variability, unit dependence and test assumptions matter; changing them needs new model-specific planning. Per-study cost is different from total hypothetical work over all simulation trials. No actual billing, pupil study or guaranteed performance is supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**
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> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **power** and resource tradeoffs **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **power** and resource tradeoffs.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Power** and resource tradeoffs. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Power** and resource tradeoffs. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- One hundred alternative trials detect 70 at n 30 and85 at n 60; unit cost 2. Compare.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task’s inputs,rule and response together.

Complete task and evidence

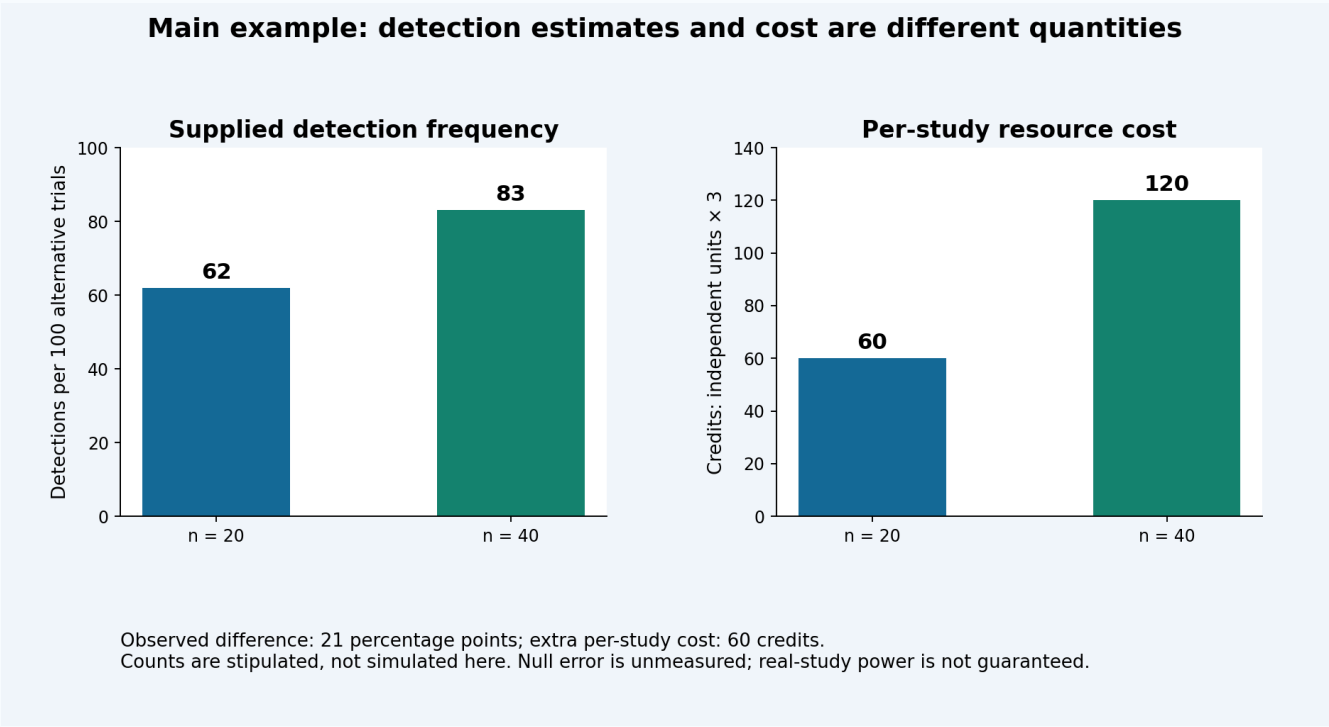
- A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

- There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

- Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/23

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/23

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are 60/100=60% and 80/100=80%.



2. Calculate design cost

n=20 costs 40 resource units; n=40 costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.
- Which labelled step matches this detail? Doubling cost adds 20 percentage points of simulated detection under these assumptions.
- 1. Compare the trade-off
- 2. Estimate simulated detection
- 3. Calculate design cost
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare the trade-off Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Changed-case question and optional hint

- Changed-case practice: Change the assumed effect or variability but keep the sample sizes. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The detection fractions need new simulation; the resource costs stay 40 and 80.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The detection fractions need new simulation; the resource costs stay 40 and 80.

Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Repair and retain the record

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Power and resource tradeoffs: the case

AI ACADEMY / AI-7/23 · 01 · LOOK AT THE FACTS

Power and resource tradeoffs: the case

The situation

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied evidence

There are 100 simulation trials per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Explain power and resource tradeoffs using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

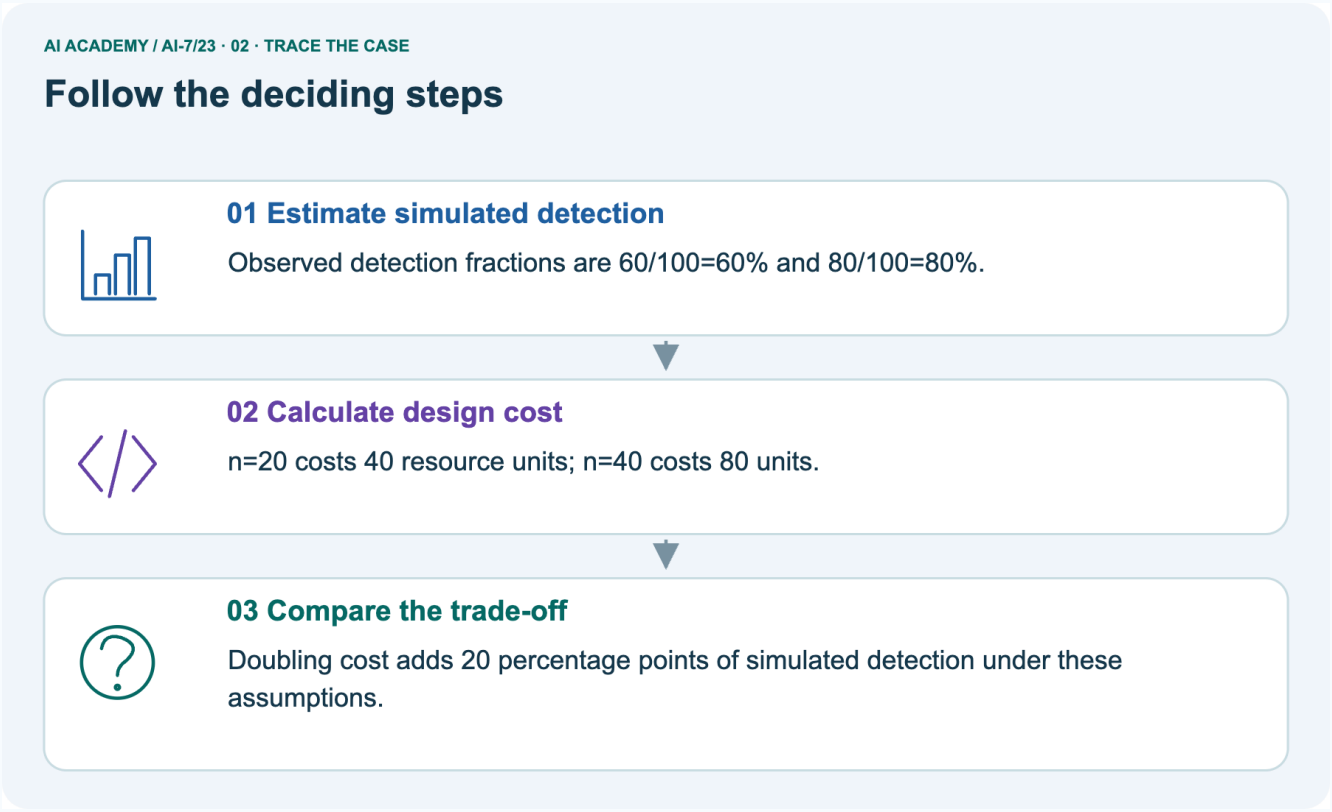
A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied facts: There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



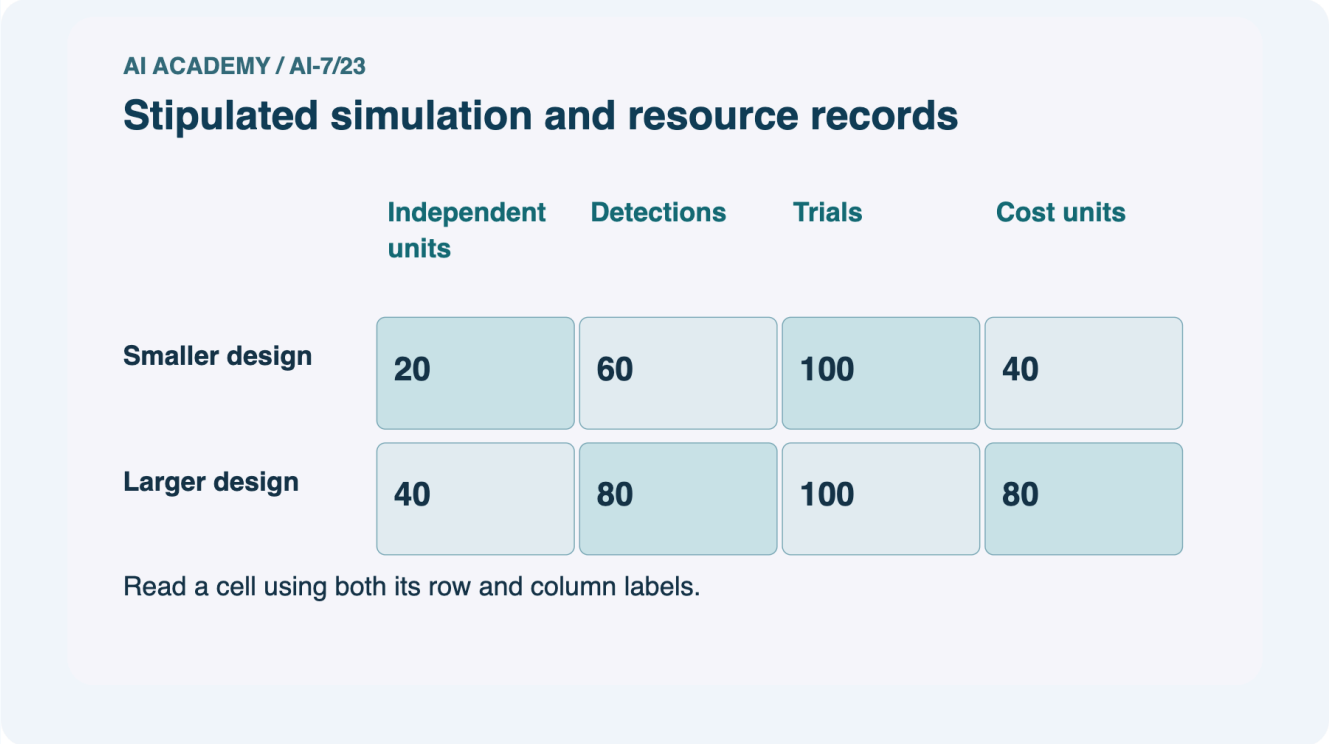
Follow the labelled picture with the matching explanation. Open the vector diagram

Estimate simulated detection: Observed detection fractions are 60/100=**60%** and 80/100=**80%**. → Calculate design cost: n=20 costs 40 resource units; n=40 costs 80 units. → Compare the trade-off: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Stipulated simulation and resource records



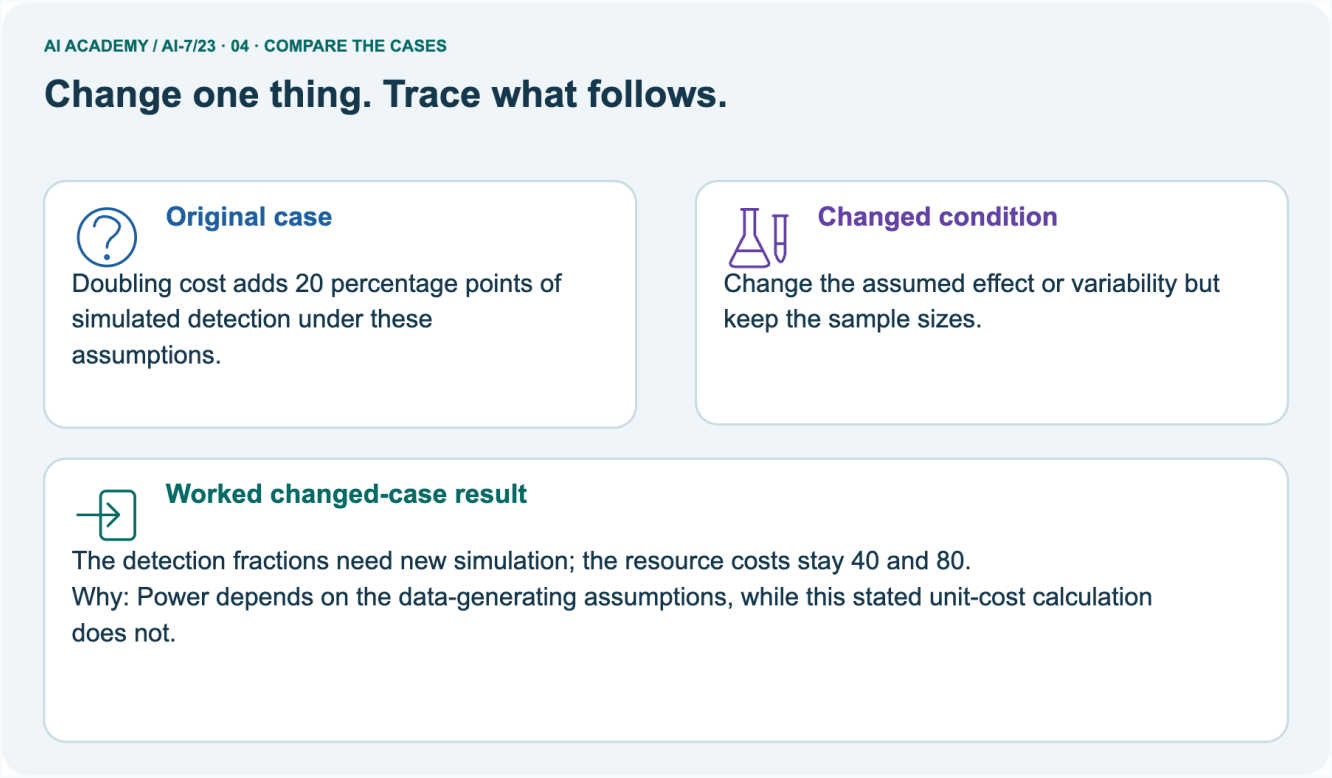
Follow the labelled picture with the matching explanation. Open the vector diagram

Stipulated simulation and resource records. Smaller design: Independent units = 20, Detections = 60, Trials = 100, Cost units = 40. Larger design: Independent units = 40, Detections = 80, Trials = 100, Cost units = 80.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

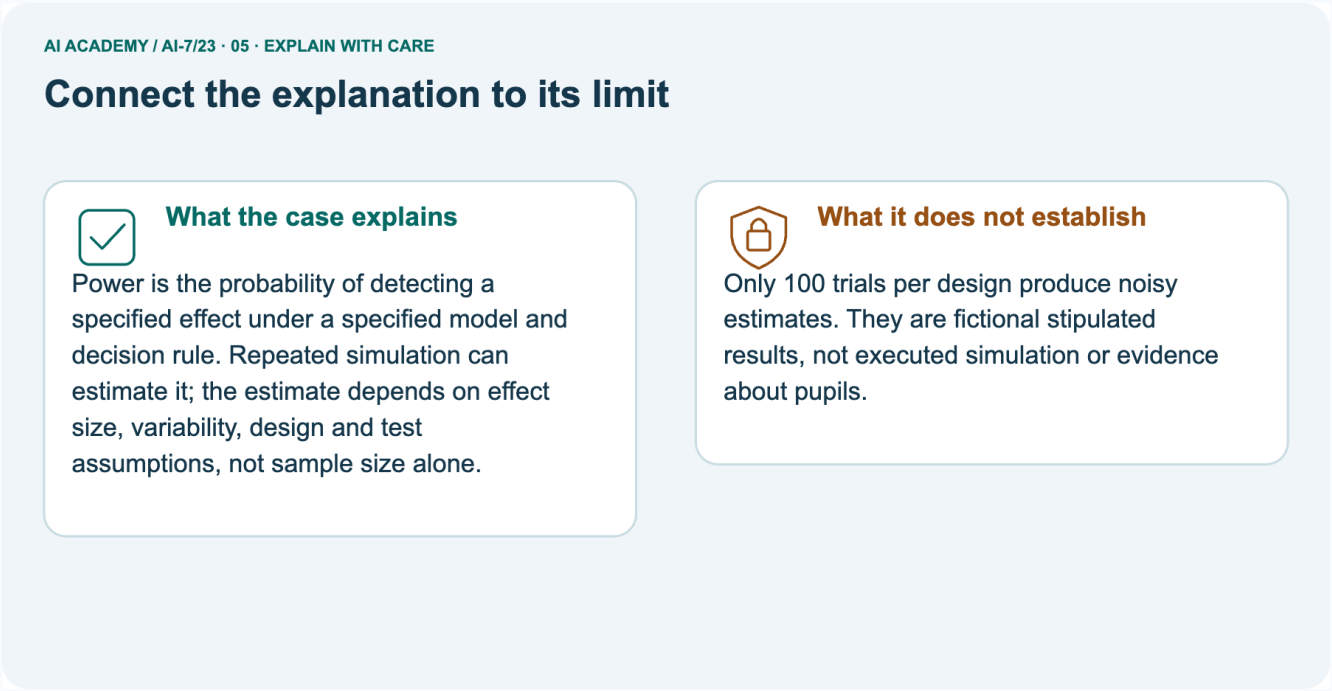
Original case: Doubling cost adds 20 percentage points of simulated detection under these assumptions. Changed condition: Change the assumed effect or variability but keep the sample sizes.

Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Power is the probability of detecting a specified effect under a specified model and decision rule.

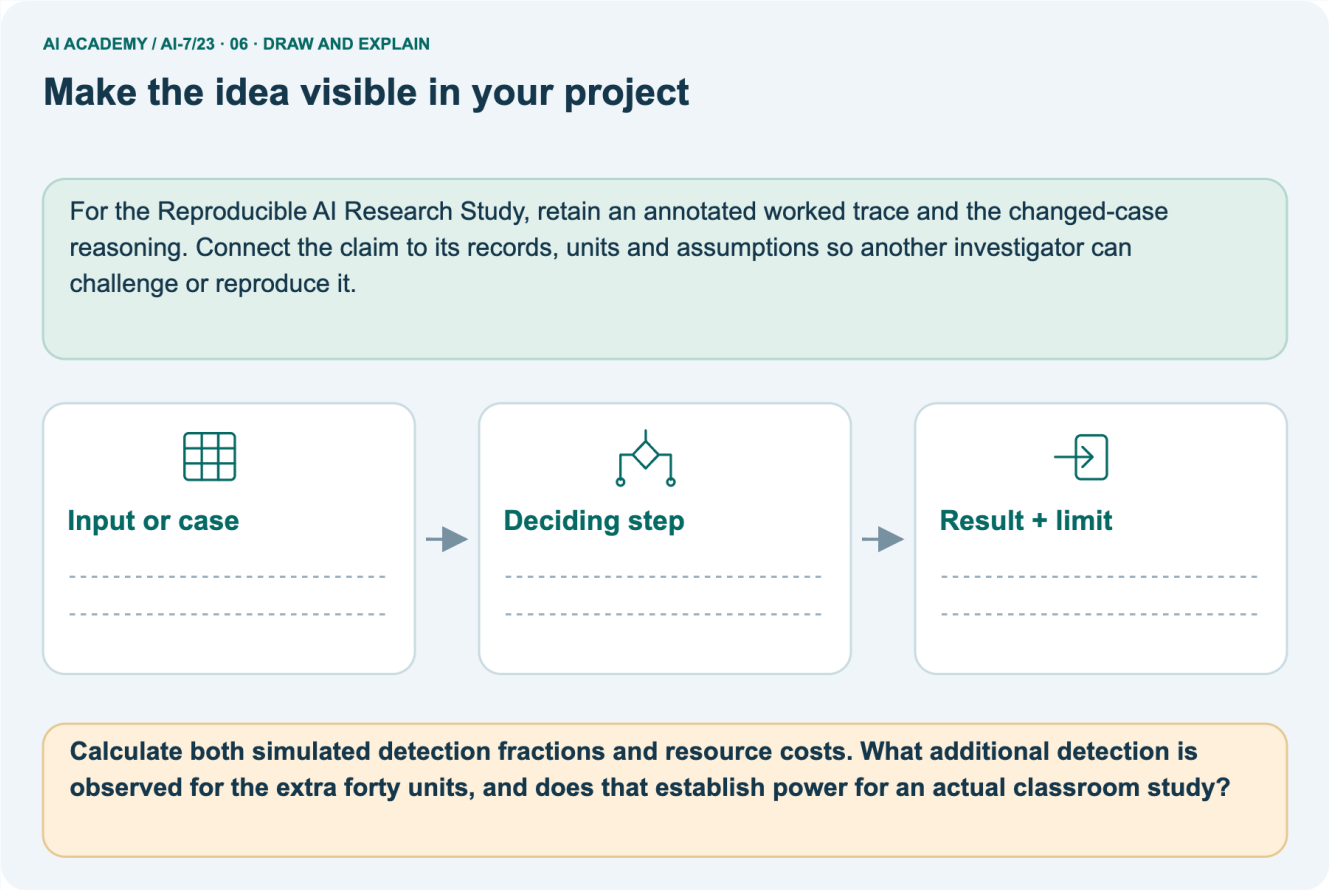
Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone. Limit: Only 100 trials per design produce noisy estimates.

They are fictional stipulated results, not executed simulation or **evidence** about pupils.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Power and resource tradeoffs: the case

02. Follow the deciding steps

03. Stipulated simulation and resource records

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.